

Curtin University, Semester 2, 2026

ECOM5005 Tutorial 9

Hypothesis Testing: Two-sample Tests

Learning Outcomes

1. Conduct hypothesis tests for the means of two independent populations.
2. Conduct hypothesis tests for the means of two related populations.
3. Conduct hypothesis tests for two population proportions.
4. Conduct hypothesis tests for the variances of two independent populations.

PART 1 – Short answer questions

Question 1

Given the following information: $S_1^2 = 4$, $n_1 = 16$; $S_2^2 = 6$, $n_2 = 25$.

(1) Calculate the degrees of freedom that should be used in the pooled-variance t test.

- A. $df = 16$ B. $df = 41$ C. $df = 25$ D. $df = 39$

(2) Calculate S_p^2 , the pooled sample variance that should be used in the pooled-variance t test.

- A. $S_p^2 = 6$ B. $S_p^2 = 5$ C. $S_p^2 = 4$ D. $S_p^2 = 5.23$

Question 2. Pooled-variance t test

A researcher randomly sampled 30 graduates of an MBA program and recorded data concerning their starting salaries. Of primary interest to the researcher was the effect of gender on starting salaries. The pooled-variance t-test output for females (Population 1) and males (Population 2) is summarised below.

(1) The researcher wants to show that female MBA graduates have a significantly lower mean starting salary than male MBA graduates. Which is the appropriate alternative hypothesis?

- A. $H_1: \mu_{\text{females}} = \mu_{\text{males}}$ B. $H_1: \mu_{\text{females}} > \mu_{\text{males}}$ C. $H_1: \mu_{\text{females}} < \mu_{\text{males}}$ D. $H_1: \mu_{\text{females}} \neq \mu_{\text{males}}$

(2) Show that the t test statistic is equal to -1.40193.

(3) What is the conclusion at the 5% significance level?

(4) What is the 90% confidence interval estimate for $\mu_{\text{females}} - \mu_{\text{males}}$?

	Females (Population 1)	Males (Population 2)
Sample size	18	12
Sample mean	\$48,266.70	\$55,000.00
Sample standard deviation	\$13,577.63	\$11,741.29

Question 3. EV fast-charging software update – paired t test

A fleet operator installs a new battery-control update on the same 12 electric vehicles. For each vehicle, engineers record the number of minutes required to add 40 kWh before and after the update.

Define $D = \text{Before} - \text{After}$, so a positive difference means the update reduces charging time.

- Explain why these data are paired rather than independent.
- At $\alpha = 0.05$, test whether the update reduces mean charging time: $H_0: \mu D \leq 0$ versus $H_1: \mu D > 0$.
- Construct a 95% confidence interval for μD and interpret it in minutes.
- State the normality assumption required for this small-sample paired t test.

EV	Before update	After update
1	31.2	29.9
2	33.5	31.7
3	29.8	29.1
4	35.1	33.0
5	32.4	30.8
6	30.6	29.7
7	34.0	31.8
8	36.2	34.1
9	31.7	30.5
10	33.1	31.4
11	29.5	28.8
12	35.4	33.2

Question 4. E-commerce checkout redesign – two population proportions

An online retailer randomly assigns website visitors to the current checkout (A) or a redesigned checkout (B). A purchase within the same session is treated as a conversion.

Current checkout A: 136 conversions from 800 visitors. Redesigned checkout B: 158 conversions from 750 visitors.

- At $\alpha = 0.05$, test whether the redesigned checkout has a higher conversion rate than the current checkout.
- Calculate the pooled proportion, the Z test statistic and the one-tail p-value.
- Construct a 95% confidence interval for $\pi_B - \pi_A$ using the unpooled standard error.
- Translate the statistical result into a decision for the product manager, and distinguish statistical from practical significance.

Question 5. Emergency-department triage systems – unequal-variance t test

A hospital compares patient waiting times under an AI-assisted triage system and the standard triage system. The samples are independent.

AI-assisted: $n_1 = 38$, $\bar{x}_1 = 41.6$ minutes, $s_1 = 12.8$ minutes. Standard: $n_2 = 34$, $\bar{x}_2 = 49.9$ minutes, $s_2 = 20.6$ minutes.

- First test $H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$ at $\alpha = 0.05$ using an F test. Put the larger sample variance in the numerator.
- Based on the variance result, select the appropriate two-sample t procedure.
- Test $H_0: \mu_1 \geq \mu_2$ versus $H_1: \mu_1 < \mu_2$ at $\alpha = 0.05$.
- Construct a 95% confidence interval for $\mu_1 - \mu_2$ and give a managerial interpretation.

PART 2 – Data exercise

Question 6. Palmer Penguins: comparing two populations in Excel

1. Use the accompanying file “Palmer_Penguins_Data.xlsx”. The data are real measurements of Adelie, Chinstrap and Gentoo penguins from the Palmer Archipelago, Antarctica. For the main analysis, compare body mass for male and female Adelie penguins.
2. (a) Open the Read_Me and Penguins_Data sheets. State the observational unit and identify the numerical response variable and grouping variable used for the main analysis.
3. (b) On Adelie_Comparison, calculate descriptive statistics for male and female body mass. Report n, mean, median, sample standard deviation, variance, minimum and maximum.
4. (c) Create a side-by-side Box & Whisker chart for male and female Adelie body mass. Comment on centre, spread and obvious outliers.
5. (d) Assuming equal population variances, test at $\alpha = 0.05$ whether male Adelie penguins have a higher mean body mass than female Adelie penguins. Use Excel Data Analysis ToolPak and report the one-tail p-value.
6. (e) Construct a 95% confidence interval for $\mu_{\text{male}} - \mu_{\text{female}}$ using the pooled-variance method and interpret it in grams.
7. (f) Test whether the male and female body-mass variances are equal using an F test at $\alpha = 0.05$. Then repeat the mean comparison using the unequal-variance (Welch) t test and compare conclusions.
8. (g) Extension – two proportions: using the Sex_Proportions sheet, test whether the recorded male proportion differs between Adelie and Gentoo penguins at $\alpha = 0.05$.
9. (h) Write a short statistical report (about 100 words) that separates statistical evidence from biological/practical interpretation.

Dataset source: Dr. Kristen Gorman and Palmer Station, Antarctica LTER; simplified palmerpenguins data. See the Read_Me sheet for citation and source links.

PART 3 – Pearson MyLab

1. Please ensure that you have finished the MyLab Homework #4.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email explaining the issue. It is important that you complete MyLab tasks on a weekly basis since the final e-test in this unit will be completed using MyLab.